Table 5 | Human Preference on Fiction Writing with the Elo Ranking System

Models	Creativity	Style	Relevance	Fluency	Overall
Weaver Ultra	**1682**	**1661**	**1689**	**1641**	**1657**
GPT-4	1507	1513	1421	1534	1508
ERNIE-Bot-4.0	1404	1409	1564	1544	1477
Gemini Pro	1513	1469	1409	1360	1430
GLM-4	1391	1445	1415	1417	1425

Table 6 | Overall Human Preference with the Elo Ranking System

Models	Creativity	Style	Relevance	Fluency	Overall
Weaver Ultra	**1589**	**1590**	**1593**	**1588**	**1576**
GLM-4	1482	1527	1491	1513	1521
GPT-4	1468	1505	1427	1501	1501
Gemini Pro	1548	1490	1434	1380	1454
ERNIE-Bot-4.0	1410	1385	1552	1515	1445

5.4. Human Evaluation

We then perform a human evaluation to compare Weaver with a few representative LLMs including GPT-4, GLM-4, ERNIE-Bot-4.0, and Gemini-pro. We recruit 44 professional Chinese writers or editors as human annotators in human evaluation. We adopt the practice in the ChatBot Arena[4] benchmark and let human annotators perform three-way pairwise comparisons between two model outputs according to their creativity, stylish, relevance, and fluency. We collect 3540 comparison results and compute the ELO rating of the compared models. The results on fiction writing and the overall comparison are shown in Table 5 and Table 6, respectively. We can see that professional writers and editors rates Weaver Ultra significantly better than compared models across all metrics. As for other compared models, we find that GPT-4 and Gemini Pro are considered to produce more creative and human-like texts compared to GLM-4 and ERNIE-Bot, we suspect this is because GLM and ERNIE are aligned using GPT distillation data, which probably harms their creativity.

5.5. User Study

A good LLM for AI-assisted writing should not only be best-performing on benchmarks but also **truly helpful** in real-world writing scenarios. To evaluate how truly helpful Weaver is, we conduct a user study where 5 professional writers are recruited as subjects. Each subject is provided with two chat interfaces, one with Weaver Ultra and the other with GPT-4. We then let each subject write two short stories (with two carefully selected topic) of around 6,000 words with two same chat interfaces powered by GPT-4 and Weaver Ultra respectively[5]. We measure the time used by the same writer for finishing the two stories and ask a professional editor to judge their quality. We find that compared to GPT-4, Weaver Ultra improves the efficiency of the writer by around 3 times. Furthermore, out of 5 topics, the human editor prefer Weaver generated story for 4 times and can not decide the winner for the remaining topic. Our user interview reveals that the efficiency improvement mainly comes from the fact that Weaver is faster and generates more human-like texts that require less

[4] `https://chat.lmsys.org/`

[5] To ensure fair comparison, we give enough time and trials for the writers to get familiar with the interface and the models.

post-editing.

6. Introducing WawaWriter

In this section, we describe WawaWriter, a next-generation AI-assisted writing platform we build to fully unleash the capabilities of Weaver. WawaWriter integrates key features of recent AI-assisted writing platforms (e.g., Notion AI) including AI-assisted generation, polishment, and summarization while also implementing a few new innovations for next-generation AI-writing experience. We describe these innovations in the following sections.

6.1. Human-AI Collaborative Writing

One major innovation in WawaWriter is a new interface for human-AI collaborative writing, which delivers a drastically different user experience compared to traditional AI-assisted writing platforms. Thanks to the Agents (Zhou et al., 2023b) framework, we are able to build controllable writing agents that act like independent human collaborators/co-authors in standard collaborative editors such as Google Docs or Notion. The writing agents understands the goal of the current document by reading customized settings such as the title or a short description of the document. It then takes actions according to the current content in the document and the recent actions of human users (or other writing agents) that reveal their focus. Human users can also chat with the writing agents in a chat interface to instruct them what to do. The ability of writing agents to use both external APIs such as web search and build-in editor APIs such as bolding or adjusting the line space enables them to accomplish tasks much more complex than what conventional AI assistants can do. With the human-agent interaction feature in the Agents framework, WriteBench also supports collaborative editing between multiple human writers and language agents. Users can customize their multiple writing agents and collaborate with one or a few of them when writing stories or articles. Users can specify tasks for each writing agent while multiple writing agents can also communicate with each other to autonomously distribute labors.

6.2. Integration of External Knowledge and Tools

Another new feature of WawaWriter is that users can now build their own personal knowledge bases via document uploading or saving web pages. WawaWriter automatically organizes and summarizes the knowledge base and then uses them as references when writing stories and articles. Specifically, we prompt an LLM to split documents into chunks based on their semantics, embed them with our embedding model, and store them in a VectorDB. During writing, we dynamically retrieve the entries of the user's personal knowledge base using semantic search using the current context in the user's editor as the query. Following Socratic Models (Zeng et al., 2023), our knowledge base also supports images in documents by using GPT-4V to do detailed captioning for each image and then using the captions as entries representing the corresponding images. Users can also edit the documents in their personal knowledge bases using all AI-writing features in WawaWriter. In addition, writing agents described in the previous section can also access the personal knowledge base of a user through function calling.

6.3. Personalized Writing Assistance

Different from current AI-assisted writing systems, WawaWriter provides personalized writing assistance for different users that suits their writing styles and content preferences. To achieve, we maintain a text-based user profile for each user which describes some basic writing habits and styles

(e.g., choice of words and punctuation, preference for the length of sentences, etc.) of the user. The user profile is periodically updated using an LLM according to the recent texts written by the user with a carefully a designed prompt. The user profile is then used as a prefix in the prompt for WEAVER. In addition to text-based user profiles, we also retrieve paragraphs that are most similar to the current context in the editor and use them as references for RAG.

6.4. Infinite Long Text Generation

WAWAWRITER also supports infinite long text generation since WEAVER natively supports the recurrent prompting technique proposed by (Zhou et al., 2023a). Specifically, to generate a very long text, we iteratively prompt WEAVER to generate an outline based on the current context and then generate a paragraph of text based on the generated outline. WAWAWRITER integrates the "step-by-step" mode and the "continuous" mode in RecurrentGPT, where the next outline is either manually selected by the user or automatically selected by an LLM. As discussed in Zhou et al. (2023a), this recurrent prompting mechanism drastically improves the creativity, consistency, and relevance of the generated long text, this is especially helpful for story/fiction writing with WAWAWRITER.

7. Discussion

In this technical report, we introduce WEAVER, a family of LLMs specialized for writing endeavors. WEAVER is continually pre-trained on carefully curated datasets and then aligned to the preferences of professional writers and editors using a novel data synthesis framework. We also release WRITEBENCH, the first benchmark for evaluating the writing capabilies of LLMs. WRITEBENCH covers multiple domains and tasks related to writing. We compare WEAVER with 10+ popular generalist LLMs and find that WEAVER ULTRA is the current state-of-the-art on the benchmark. Our user study also confirms the superiority of WEAVER in real-world AI-assisted writing scenarios. The results also confirm the effectiveness of our data synthesis pipeline for training domain-specific LLMs.

References

J. Ainslie, J. Lee-Thorp, M. de Jong, Y. Zemlyanskiy, F. Lebrón, and S. Sanghai. Gqa: Training generalized multi-query transformer models from multi-head checkpoints. arXiv preprint arXiv:2305.13245, 2023.

Anthropic. Introducing Claude, 2023. URL https://www.anthropic.com/index/introducing-claude.

Y. Bai, S. Kadavath, S. Kundu, A. Askell, J. Kernion, A. Jones, A. Chen, A. Goldie, A. Mirhoseini, C. McKinnon, C. Chen, C. Olsson, C. Olah, D. Hernandez, D. Drain, D. Ganguli, D. Li, E. Tran-Johnson, E. Perez, J. Kerr, J. Mueller, J. Ladish, J. Landau, K. Ndousse, K. Lukosuite, L. Lovitt, M. Sellitto, N. Elhage, N. Schiefer, N. Mercado, N. DasSarma, R. Lasenby, R. Larson, S. Ringer, S. Johnston, S. Kravec, S. E. Showk, S. Fort, T. Lanham, T. Telleen-Lawton, T. Conerly, T. Henighan, T. Hume, S. R. Bowman, Z. Hatfield-Dodds, B. Mann, D. Amodei, N. Joseph, S. McCandlish, T. Brown, and J. Kaplan. Constitutional ai: Harmlessness from ai feedback, 2022.

T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. Henighan, R. Child, A. Ramesh, D. M. Ziegler, J. Wu, C. Winter, C. Hesse, M. Chen, E. Sigler, M. Litwin, S. Gray, B. Chess, J. Clark, C. Berner, S. McCandlish, A. Radford, I. Sutskever, and D. Amodei. Language models are few-shot learners, 2020.